

"Continuity"

CH- 23

MATH $\rightarrow$ H.

Continuity:

A function f is said to be continuous at a point c , if the following conditions are satisfied -

- i) $f(c)$ is defined
- ii) $\lim_{x \rightarrow c} f(x)$ exists
- iii) $\lim_{x \rightarrow c} f(x) = f(c)$

If one or more of the conditions in this definition fails to hold, then we say, that f has a discontinuity at the point c .

If f is continuous at each point of an open interval (a, b) , then we will say that f is continuous on (a, b) . If f is continuous on $(-\infty, \infty)$ then we will say that f is continuous everywhere.

Example: Determine whether the following functions are continuous at $x=2$.

(a) $f(x) = \frac{x^2-4}{x-2}$

(b) $g(x) = \begin{cases} \frac{x^2-4}{x-2}, & x \neq 2 \\ 3, & x = 2 \end{cases}$

(c) $h(x) = \begin{cases} \frac{x^2-4}{x-2}, & x \neq 2 \\ 4, & x = 2 \end{cases}$

Solⁿ

(a) The function f is undefined at $x=2$.

Hence the function is not continuous at $x=2$.

(b) $\lim_{x \rightarrow 2} g(x) = \lim_{x \rightarrow 2} \frac{x^2-4}{x-2} = \lim_{x \rightarrow 2} x+2 = 4$

The function g is defined at $x=2$ but its value there is $g(2)=3$, which is not the same as the limit at the point.

Hence, g is not continuous at $x=2$.

(c) $\lim_{x \rightarrow 2} h(x) = \lim_{x \rightarrow 2} \frac{x^2-4}{x-2} = \lim_{x \rightarrow 2} x+2 = 4$

The value of the function h at $x=2$ is $h(2)=4$, which is the same as the limit at that point, hence h is continuous at $x=2$.

e.g. 2 Does $f(x) = \sqrt{9-x^2}$ continuous?

Solve:

The natural domain of the given function is $[-3, 3]$. We need to investigate the continuity of f on $(-3, 3)$ and at the end points.

If c is any point in the interval $(-3, 3)$, then

$$\begin{aligned}\lim_{x \rightarrow c} f(x) &= \lim_{x \rightarrow c} \sqrt{9-x^2} \\ &= \sqrt{\lim_{x \rightarrow c} 9-x^2} \\ &= \sqrt{9-c^2} \\ &= f(c)\end{aligned}$$

which shows that f is continuous on each point of $(-3, 3)$.

$$\text{Now, } \lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^-} \sqrt{9-x^2} = \sqrt{\lim_{x \rightarrow 3^-} 9-x^2} = 0 = f(3) \quad (= f(3))$$

$$\lim_{x \rightarrow -3^+} f(x) = \lim_{x \rightarrow -3^+} \sqrt{9-x^2} = \sqrt{\lim_{x \rightarrow -3^+} (9-x^2)} = 0 = f(-3)$$

The function is continuous on the closed interval $[-3, 3]$.

Removable discontinuity:

A function f is said to have

a R.D at $x=c$, if $\lim_{x \rightarrow c} f(x)$ exists but

$$f(c) \neq \lim_{x \rightarrow c} f(x)$$

either because $f(c)$ is undefined or the value of $f(c)$ differs from the values of the limits.

Example 1:- Find the discontinuity if any of the function

$$f(x) = \begin{cases} \frac{3}{x-1} & , x \neq 1 \\ 3 & , x = 1 \end{cases}$$

Solⁿ Here given $f(1) = 3$

$$\text{Now, } \lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} \frac{3}{x-1} = +\infty$$

$$\text{and } \lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} \frac{3}{x-1} = -\infty$$

Hence $\lim_{x \rightarrow 1} f(x)$ does not exist

Hence the given function has discontinuity at $x = 1$.

Ex 12

A function $f(x)$ is defined as follows:

$$f(x) = \begin{cases} 3 - 2x, & -3/2 \leq x < 0 \\ 3 + 2x, & 0 \leq x < 3/2 \\ -3 - 2x, & x \geq 3/2 \end{cases}$$

Show that the function is continuous at $x=0$ and discontinuous at $x=3/2$.

Solⁿ

Consider at $x=0$

Now,

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} 3 - 2x = 3$$

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} 3 + 2x = 3$$

$$\text{and } f(0) = 3 - 0 = 3$$

$$\therefore f(0) = \lim_{x \rightarrow 0} f(x)$$

Therefore the function is continuous at $x=0$.

Consider at $x=3/2$

$$\text{Now, } \lim_{x \rightarrow 3/2^+} f(x) = \lim_{x \rightarrow 3/2^+} -3 - 2x = -6$$

$$\lim_{x \rightarrow 3/2^-} f(x) = \lim_{x \rightarrow 3/2^-} 3 - 2x = 0$$

$$\text{and } f(3/2) = -6$$

$$\therefore f(3/2) \neq \lim_{x \rightarrow 3/2} f(x)$$

$\therefore$ The function is discontinuous at $x = 3/2$.

Ex-3

Find the points of discontinuity and determine whether the discontinuities are removable.

(a) $f(x) = \frac{x-2}{|x|-2}$

(b) $f(x) = \frac{x^2+3x}{x+3}$

Solⁿ-a

The given function can be expressed as

$$f(x) = \begin{cases} \frac{x-2}{x-2} & \text{if } x > 0 \\ \frac{x-2}{-x-2} & \text{if } x < 0 \\ \frac{x-2}{-2} & \text{if } x = 0 \end{cases}$$

The function is undefined at $x=2$ and $x=-2$ considering $x=2$

$$\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} \frac{x-2}{x-2} = 1$$

$$\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} \frac{x-2}{x-2} = 1$$

But $f(2)$ is undefined

$\therefore$ The function has a removable discontinuity at the point ~~of~~ of $x=2$.